

I evaluated five different encoding methods to assess their accuracy. Below are the detailed results for reference.

```
(venv) PS C:\ARSHETTY\QUANTUM\EVERYTHING_SUMMERISED\RETP2> python start.py --data_dir C:\ARSHETTY\QUANTUM\EVERYTHING_SUMMERISED\Final_workspace\DATASET --steps 4
=====
IoT IDS Pipeline -- Classical + Quantum Hybrid Autoencoder
=====

Data dir      : C:\ARSHETTY\QUANTUM\EVERYTHING_SUMMERISED\Final_workspace\DATASET
Output dir    : C:\ARSHETTY\QUANTUM\EVERYTHING_SUMMERISED\RETP2\OUTPUT
Steps to run  : [4]
Label column  : label
Benign class  : BenignTraffic
Max/class    : 8,000
K features    : 8
Zero-day fams : ['DDoS-HTTP_Flood', 'Backdoor_Malware', 'Uploading_Attack', 'VulnerabilityScan']

Found 169 CSV file(s) in data_dir.
=====

RUNNING Step 4 - Basic QAE ...
=====

ENCODING COMPARISON STUDY FOR QUANTUM AUTOENCODER
Comparing 5 encoding methods on the same anomaly detection task
=====

[DATA] Train: (5600, 8) | Val: (1200, 8) | Test: (1200, 8) | Known: (214621, 8) | ZD: (20470, 8)
```

Dense Angle:

```
=====
PIPELINE: DENSE_ANGLE
Dense Angle Encoding (Re-uploading, 2 features/qubit)
Qubits: 4 | Latent: 2 | Trash: 2
Params: 12
=====

[INFO] Circuit built: 4 qubits + 1 ancilla = 5 wires
[INFO] Trainable parameters: 12

[TRAIN] Encoding: dense_angle
[TRAIN] Parameters: 12 | Qubits: 4
[TRAIN] Optimizer: COBYLA | Epochs: 50 | Batch: 32
  Epoch 10/50 | Loss: 0.3294 | Best: 0.2679 | Time: 439.2s
  Epoch 20/50 | Loss: 0.3879 | Best: 0.2679 | Time: 871.4s
  Epoch 30/50 | Loss: 0.3942 | Best: 0.2679 | Time: 1300.5s
  Epoch 40/50 | Loss: 0.3520 | Best: 0.2488 | Time: 1752.5s
  Epoch 50/50 | Loss: 0.3899 | Best: 0.2488 | Time: 2226.6s
[TRAIN] Complete in 2226.6s | Best loss: 0.2488
[SAVED] Parameters -> C:\ARSHETTY\QUANTUM\EVERYTHING_SUMMERISED\RETP2\OUTPUT\QUANTUM_RESULTS\theta_dense_angle.npy

[SCORE] Computing anomaly scores...
  Scoring validation benign (n=500)...
  Scored 500/500 samples...
  Scoring test benign (n=500)...
  Scored 500/500 samples...
  Scoring zero-day attacks (n=500)...
  Scored 500/500 samples...
[Threshold] tau = 0.9443 | Target FAR: 5% | Actual val FAR: 5.00%

=====
Results: dense_angle
=====

ZDDR (Zero-Day Detection Rate) : 49.20%
FAR (False Alarm Rate)         : 6.20%
Precision                       : 0.8881
Recall (= ZDDR)                : 0.4920
F1 Score                       : 0.6332
AUC-ROC                        : 0.7121
Threshold tau                   : 0.9443
=====
```

Angle encoding:

```
PIPELINE: ANGLE
Angle Encoding (Repeated, 1 feature/qubit)
Qubits: 8 | Latent: 4 | Trash: 4
Params: 24
=====
[INFO] Circuit built: 8 qubits + 1 ancilla = 9 wires
[INFO] Trainable parameters: 24

[TRAIN] Encoding: angle
[TRAIN] Parameters: 24 | Qubits: 8
[TRAIN] Optimizer: COBYLA | Epochs: 50 | Batch: 32
  Epoch 10/50 | Loss: 0.4542 | Best: 0.3959 | Time: 690.6s
  Epoch 20/50 | Loss: 0.4219 | Best: 0.3705 | Time: 989.3s
  Epoch 30/50 | Loss: 0.3388 | Best: 0.3388 | Time: 1295.0s
  Epoch 40/50 | Loss: 0.4379 | Best: 0.3388 | Time: 1639.6s
  Epoch 50/50 | Loss: 0.4520 | Best: 0.3346 | Time: 1966.8s
[TRAIN] Complete in 1966.8s | Best loss: 0.3346
[SAVED] Parameters -> C:\ARSHETTY\QUANTUM\EVERYTHING_SUMMERISED\RETP2\OUTPUT\QUANTUM_RESULTS\theta_angle.npy

[SCORE] Computing anomaly scores...
Scoring validation benign (n=500)...
Scored 500/500 samples...
Scoring test benign (n=500)...
Scored 500/500 samples...
Scoring zero-day attacks (n=500)...
Scored 500/500 samples...
[Threshold] tau = 0.9261 | Target FAR: 5% | Actual val FAR: 5.00%

=====
Results: angle
=====
ZDDR (Zero-Day Detection Rate) : 8.60%
FAR (False Alarm Rate)       : 5.60%
Precision                     : 0.6056
Recall (= ZDDR)              : 0.0860
F1 Score                     : 0.1506
AUC-ROC                      : 0.4611
Threshold tau                 : 0.9261
=====
```

Amplitude:

```
=====
PIPELINE: AMPLITUDE
Amplitude Encoding (8 features -> 2^3 amplitudes)
Qubits: 3 | Latent: 2 | Trash: 1
Params: 9
=====

[INFO] Circuit built: 3 qubits + 1 ancilla = 4 wires
[INFO] Trainable parameters: 9

[TRAIN] Encoding: amplitude
[TRAIN] Parameters: 9 | Qubits: 3
[TRAIN] Optimizer: COBYLA | Epochs: 50 | Batch: 32
  Epoch 10/50 | Loss: 0.0241 | Best: 0.0194 | Time: 83.8s
  Epoch 20/50 | Loss: 0.0265 | Best: 0.0188 | Time: 176.6s
  Epoch 30/50 | Loss: 0.0264 | Best: 0.0171 | Time: 261.3s
  Epoch 40/50 | Loss: 0.0221 | Best: 0.0169 | Time: 345.6s
  Epoch 50/50 | Loss: 0.0260 | Best: 0.0153 | Time: 432.9s
[TRAIN] Complete in 432.9s | Best loss: 0.0153
[SAVED] Parameters -> C:\ARSHETTY\QUANTUM\EVERYTHING_SUMMERISED\RETP2\OUTPUT\QUANTUM_RESULTS\theta_amplitude.npy

[SCORE] Computing anomaly scores...
  Scoring validation benign (n=500)...
  Scored 500/500 samples...
  Scoring test benign (n=500)...
  Scored 500/500 samples...
  Scoring zero-day attacks (n=500)...
  Scored 500/500 samples...
[Threshold] tau = 0.1147 | Target FAR: 5% | Actual val FAR: 5.00%

=====
Results: amplitude
=====
ZDDR (Zero-Day Detection Rate) : 26.40%
FAR (False Alarm Rate)       : 5.80%
Precision                     : 0.8199
Recall (= ZDDR)              : 0.2640
F1 Score                     : 0.3994
AUC-ROC                      : 0.7336
Threshold tau                 : 0.1147
=====
```

Zzfeature map:

```
PIPELINE: ZZ_FEATURE_MAP
ZZ Feature Map (Hadamard + pairwise RZZ)
Qubits: 8 | Latent: 4 | Trash: 4
Params: 24

=====
[INFO] Circuit built: 8 qubits + 1 ancilla = 9 wires
[INFO] Trainable parameters: 24

[TRAIN] Encoding: zz_feature_map
[TRAIN] Parameters: 24 | Qubits: 8
[TRAIN] Optimizer: COBYLA | Epochs: 50 | Batch: 32
  Epoch 10/50 | Loss: 0.4725 | Best: 0.4689 | Time: 493.0s
  Epoch 20/50 | Loss: 0.4639 | Best: 0.4402 | Time: 1003.1s
  Epoch 30/50 | Loss: 0.4689 | Best: 0.4402 | Time: 1509.8s
  Epoch 40/50 | Loss: 0.4671 | Best: 0.4402 | Time: 2010.1s
  Epoch 50/50 | Loss: 0.1613 | Best: 0.1512 | Time: 2509.6s
[TRAIN] Complete in 2509.6s | Best loss: 0.1512
[SAVED] Parameters -> C:\ARSHETTY\QUANTUM\EVERYTHING_SUMMERISED\RETP2\OUTPUT\QUANTUM_RESULTS\theta_zz_feature_map.npy

[SCORE] Computing anomaly scores...
  Scoring validation benign (n=500)...
  Scored 500/500 samples...
  Scoring test benign (n=500)...
  Scored 500/500 samples...
  Scoring zero-day attacks (n=500)...
  Scored 500/500 samples...
  [Threshold] tau = 0.5021 | Target FAR: 5% | Actual val FAR: 5.00%

=====
Results: zz_feature_map
=====
ZDDR (Zero-Day Detection Rate) : 0.00%
FAR (False Alarm Rate)         : 2.40%
Precision                       : 0.0000
Recall (= ZDDR)                : 0.0000
F1 Score                       : 0.0000
AUC-ROC                        : 0.3942
Threshold tau                   : 0.5021
=====
```

Phase:

```
PIPELINE: PHASE
Quantum Phase Encoding (RZ in superposition, bonus)
Qubits: 8 | Latent: 4 | Trash: 4
Params: 24
=====
[INFO] Circuit built: 8 qubits + 1 ancilla = 9 wires
[INFO] Trainable parameters: 24

[TRAIN] Encoding: phase
[TRAIN] Parameters: 24 | Qubits: 8
[TRAIN] Optimizer: COBYLA | Epochs: 50 | Batch: 32
  Epoch 10/50 | Loss: 0.1389 | Best: 0.1389 | Time: 338.0s
  Epoch 20/50 | Loss: 0.1468 | Best: 0.1389 | Time: 673.5s
  Epoch 30/50 | Loss: 0.1407 | Best: 0.1389 | Time: 1011.6s
  Epoch 40/50 | Loss: 0.2085 | Best: 0.1389 | Time: 1342.4s
  Epoch 50/50 | Loss: 0.1601 | Best: 0.1124 | Time: 1687.3s
[TRAIN] Complete in 1687.3s | Best loss: 0.1124
[SAVED] Parameters -> C:\ARSHETTY\QUANTUM\EVERYTHING_SUMMERISED\RETP2\OUTPUT\QUANTUM_RESULTS\theta_phase.npy

[SCORE] Computing anomaly scores...
  Scoring validation benign (n=500)...
  Scored 500/500 samples...
  Scoring test benign (n=500)...
  Scored 500/500 samples...
  Scoring zero-day attacks (n=500)...
  Scored 500/500 samples...
[Threshold] tau = 0.6390 | Target FAR: 5% | Actual val FAR: 5.00%

=====
Results: phase
=====
ZDDR (Zero-Day Detection Rate) : 0.20%
FAR (False Alarm Rate)       : 6.20%
Precision                     : 0.0312
Recall (= ZDDR)              : 0.0020
F1 Score                      : 0.0038
AUC-ROC                      : 0.2332
Threshold tau                 : 0.6390
=====
```

In 1651 Col 1 Spaces: 4 UTF-8 LF {} Python 3.11.9 (venv

Visualisation:

GENERATING VISUALIZATIONS

```
[VIZ] Classical feature heatmap...
[FIG] Saved: ./results/encoding_comparison/figures\01_classical_features_heatmap.png
```

```
[VIZ] dense_angle: statevector + histogram + circuit...
[FIG] Saved: ./results/encoding_comparison/figures\02_statevector_dense_angle.png
[FIG] Saved: ./results/encoding_comparison/figures\03_measurement_dense_angle.png
```

Circuit Diagram for 'dense_angle' (4 qubits):

```
-----
0: --RY(3.10)--RY(2.19)-+?-----+X--RY(3.10)--RY(2.19)-+ +Probs
1: --RY(2.56)--RY(3.04)+X-+?----|---RY(2.56)--RY(3.04)-+ +Probs
2: --RY(3.05)--RY(1.05)----+X-+?-|---RY(3.05)--RY(1.05)-+ +Probs
3: --RY(1.63)--RY(2.90)-----+X-+?-RY(1.63)--RY(2.90)-+ +Probs
-----
```

```
[FIG] Saved circuit diagram: ./results/encoding_comparison/figures\04_circuit_dense_angle.txt
```

```
[VIZ] angle: statevector + histogram + circuit...
[FIG] Saved: ./results/encoding_comparison/figures\02_statevector_angle.png
[FIG] Saved: ./results/encoding_comparison/figures\03_measurement_angle.png
```

Circuit Diagram for 'angle' (8 qubits):

```
-----
0: --RY(1.47)-+?--RY(1.47)-----+ +Probs
1: --RY(1.62)+X-+?-----RY(1.62)-----+ +Probs
2: --RY(0.15)----+X-----+?-----RY(0.15)-----+ +Probs
3: --RY(1.75)-----+X-----+?-----RY(1.75)-----+ +Probs
4: --RY(0.39)-----+X-----+?-----RY(0.39)-----+ +Probs
5: --RY(0.83)-----+X-----+?-----RY(0.83)-----+ +Probs
6: --RY(2.53)-----+X-----+?-----RY(2.53)-+ +Probs
7: --RY(1.59)-----+X-----RY(1.59)-+ +Probs
-----
```

```
[FIG] Saved circuit diagram: ./results/encoding_comparison/figures\04_circuit_angle.txt
```

```
[VIZ] amplitude: statevector + histogram + circuit...
[FIG] Saved: ./results/encoding_comparison/figures\02_statevector_amplitude.png
[FIG] Saved: ./results/encoding_comparison/figures\03_measurement_amplitude.png
```

Circuit Diagram for 'amplitude' (3 qubits):

```
-----
0: -+|?>-+ +Probs
1: -+|?>-+ +Probs
2: -+|?>-+ +Probs
-----
```

[FIG] Saved circuit diagram: ./results/encoding_comparison/figures\04_circuit_amplitude.txt

```
[VIZ] zz_feature_map: statevector + histogram + circuit...
[FIG] Saved: ./results/encoding_comparison/figures\02_statevector_zz_feature_map.png
[FIG] Saved: ./results/encoding_comparison/figures\03_measurement_zz_feature_map.png
```

Circuit Diagram for 'zz_feature_map' (8 qubits):

```
-----
0: --H--RZ(1.71)-+?-----+?--H--RZ(1.71)-----+?-----+?----- ???
1: --H--RZ(1.52)+X--RZ(1.30)+X-+?-----+?--H--RZ(1.52)+X--RZ(1.30)+X-----+? ???
2: --H--RZ(1.44)-----+X--RZ(1.10)+X-+?-----+?--H-----RZ(1.44)+X ???
3: --H--RZ(1.13)-----+X--RZ(0.82)+X-+?-----+?--H-----RZ(1.13)+X ???
4: --H--RZ(0.66)-----+X-----RZ(0.37)+X ???
5: --H--RZ(0.87)----- ???
6: --H--RZ(0.12)----- ???
7: --H--RZ(0.18)----- ???

0: ??? ----- ???
1: ??? -----+?----- ???
2: ??? --RZ(1.10)+X-----+?-----+?----- ???
3: ??? --H-----RZ(1.13)+X--RZ(0.82)+X-----+?-----+?----- ???
4: ??? -+?-----+?--H-----RZ(0.66)+X--RZ(0.37)+X-----+?-----+?----- ???
5: ??? +X-----RZ(0.28)+X-+?-----+?--H-----RZ(0.87)+X--RZ(0.28)+X----- ???
6: ??? -----+X-----RZ(0.05)+X-+?-----+?--H-----RZ(0.12) ???
7: ??? -----+X-----RZ(0.01)+X--H-----RZ(0.18) ???
```

```

0: ??? -----+ +Probs
1: ??? -----+ +Probs
2: ??? -----+ +Probs
3: ??? -----+ +Probs
4: ??? -----+ +Probs
5: ??? -+?-----+?-----+ +Probs
6: ??? -+X--RZ(0.05)-+X-+?-----+?--+ +Probs
7: ??? -----+X--RZ(0.01)-+X-+ +Probs

```

[FIG] Saved circuit diagram: ./results/encoding_comparison/figures\04_circuit_zz_feature_map.txt

[VIZ] phase: statevector + histogram + circuit...

[FIG] Saved: ./results/encoding_comparison/figures\02_statevector_phase.png

[FIG] Saved: ./results/encoding_comparison/figures\03_measurement_phase.png

Circuit Diagram for 'phase' (8 qubits):

```

0: --H--RZ(1.10)-+?--RZ(1.10)--H-----+ ???
1: --H--RZ(1.19)+X-+?-----RZ(1.19)--H-----+ ???
2: --H--RZ(2.50)-----+X-----+?-----RZ(2.50)--H-----+ ???
3: --H--RZ(2.15)-----+X-----+?-----RZ(2.15)--H-----+ ???
4: --H--RZ(2.56)-----+X-----+?-----RZ(2.56)--H-----+ ???
5: --H--RZ(1.15)-----+X-----+?-----RZ(1.15)--H-----+ ???
6: --H--RZ(2.59)-----+X-----+?-----RZ(2.59)--H-+ ???
7: --H--RZ(2.98)-----+X-----+X-----RZ(2.98)--H-+ ???

```

```

0: ??? +Probs
1: ??? +Probs
2: ??? +Probs
3: ??? +Probs
4: ??? +Probs
5: ??? +Probs
6: ??? +Probs
7: ??? +Probs

```

[FIG] Saved circuit diagram: ./results/encoding_comparison/figures\04_circuit_phase.txt

[FIG] Saved: ./results/encoding_comparison/figures\05_training_loss_curves.png

[FIG] Saved: ./results/encoding_comparison/figures\06_metrics_comparison.png

[FIG] Saved: ./results/encoding_comparison/figures\07_qubit_vs_accuracy.png

[FIG] Saved: ./results/encoding_comparison/figures\08_far_comparison.png

Encoding	Qubits	Params	ZDDR	FAR	Prec	Recall	F1	AUC
dense_angle	4	12	0.4920	0.0620	0.8881	0.4920	0.6332	0.7121
angle	8	24	0.0860	0.0560	0.6056	0.0860	0.1506	0.4611
amplitude	3	9	0.2640	0.0580	0.8199	0.2640	0.3994	0.7336
zz_feature_map	8	24	0.0000	0.0240	0.0000	0.0000	0.0000	0.3942
phase	8	24	0.0020	0.0620	0.0312	0.0020	0.0038	0.2332

Encoding	Qubits	Params	ZDDR	FAR	Prec	Recall	F1	AUC
dense_angle	4	12	0.4920	0.0620	0.8881	0.4920	0.6332	0.7121
angle	8	24	0.0860	0.0560	0.6056	0.0860	0.1506	0.4611
amplitude	3	9	0.2640	0.0580	0.8199	0.2640	0.3994	0.7336
zz_feature_map	8	24	0.0000	0.0240	0.0000	0.0000	0.0000	0.3942
phase	8	24	0.0020	0.0620	0.0312	0.0020	0.0038	0.2332

ENCODING METHOD PROS & CONS SUMMARY

```
Qubits: 4 | Latent: 2 | Trash: 2
```

Trainable params (RealAmplitudes)

PROS:

 $+ Q$

- + Re-uploading creates rich feature interactions
- + Well-suited for NISQ devices with limited qubits
- + Proven baseline in QAE literature

CONS:

- Fe

- Sequential RY rotations on same qubit can interfere
- CNOT ring entanglement adds depth for every encoding layer

Qubits: 8 | Latent: 4 | Trash: 4

Trainable params (RealAmplitudes)

PROS:

+ S:

- + No information compression at encoding stage
- + Easy to interpret and debug
- + Repeated structure helps learn correlations

```

[ZZ_FEATURE_MAP] - ZZ Feature Map (Hadamard + pairwise RZZ)
Qubits: 8 | Latent: 4 | Trash: 4
Trainable params (RealAmplitudes x3): 24
PROS:
+ Captures second-order feature interactions ( $x_i * x_j$ )
+ Proven advantage in quantum kernel methods
+ Classically hard to simulate (potential quantum advantage)
+ Well-studied in QML literature (Havlíček et al., 2019)
CONS:
- Qubit-expensive: needs 8 qubits for 8 features
- Deep circuit: pairwise RZZ gates add significant depth
- May over-capture interactions for simple datasets
- Training can be harder due to barren plateaus

[PHASE] - Quantum Phase Encoding (RZ in superposition, bonus)
Qubits: 8 | Latent: 4 | Trash: 4
Trainable params (RealAmplitudes x3): 24
PROS:
+ Phase encoding preserves magnitude information better than RY
+ Hadamard + RZ creates rich interference patterns
+ Re-uploading with phases enables complex feature mixing
+ Final Hadamard converts phase info to measurable amplitudes
CONS:
- Qubit-expensive: needs 8 qubits for 8 features
- Phase encoding is less intuitive than amplitude encoding
- More circuit depth than simple angle encoding
- Performance sensitive to feature scaling/normalization

=====

[SAVED] Results -> C:\ARSHETTY\QUANTUM\EVERYTHING_SUMMERISED\RETP2\OUTPUT\QUANTUM_RESULTS\encoding_comparison_results.json

=====

STUDY COMPLETE
Results saved to: C:\ARSHETTY\QUANTUM\EVERYTHING_SUMMERISED\RETP2\OUTPUT\QUANTUM_RESULTS/
Figures saved to: ./results/encoding_comparison/figures/

=====

ALL DONE!
=====

```

OUTPUT:









